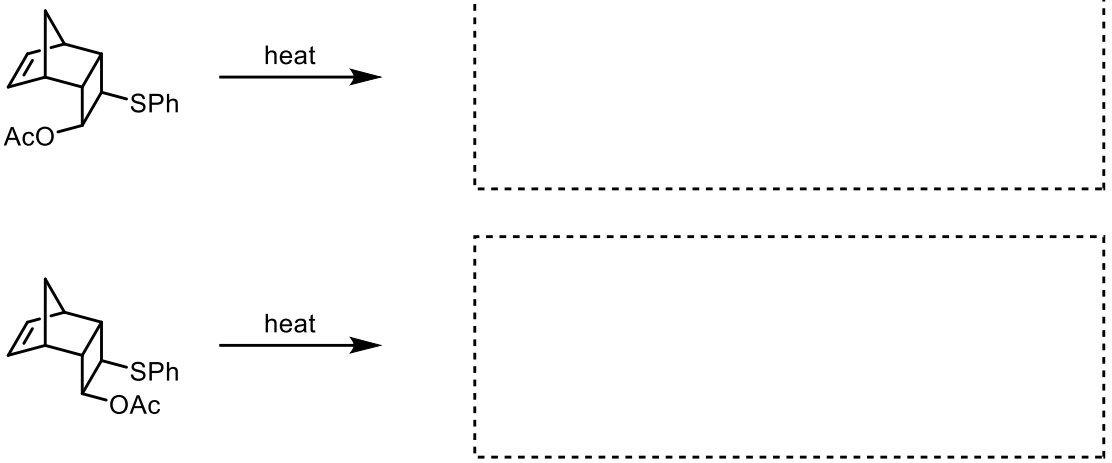
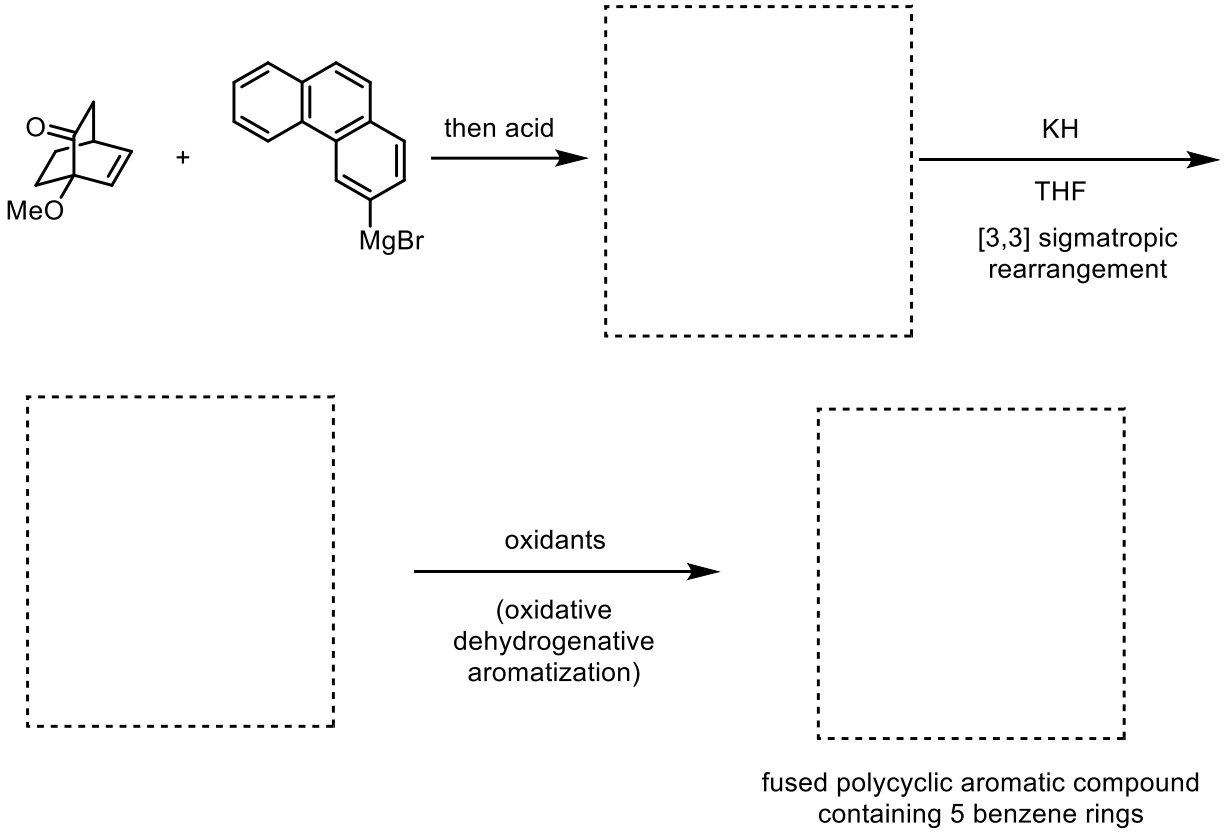
Question 1: Propose the products of the thermal reactions shown below. Explain the stereoselectivity of reach reaction. (Hint: We discussed in lecture that cyclopentadiene exists as a dimer, and the monomer can be generated at high temperature via a retro–Diels–Alder reaction. The first step of each reaction is a retro–Diels–Alder reaction)



Question 2. Propose the chemical structures of products for the following reaction.



Question 3. Proposed the reaction mechanism for the following transformation. Which product is the major product under thermal conditions? Which product is the major product under photochemical conditions?

